Mathematical overview

Stability and minimizing total rank are different objectives. In the printed complete three-by-three profile, AY/BZ/CX uniquely minimizes aggregate rank at 10 but BX blocks it. AY/BX/CZ is uniquely stable, with total 11. Rank numbers are an invented puzzle score, not a measurement of real happiness.

With strict incomplete two-side lists, only mutually listed pairs are acceptable, every acceptable partner is better than being unpaired, and a stable matching has no allowed pair both preferring one another. Perfect pairing need not exist even with equal sides. Nevertheless all stable matchings match exactly the same objects; partners can change while the unpaired set cannot. A finite alternating-path argument proves this invariant.

Removing the two-side restriction changes existence. Four objects in one group have only three perfect pairings. The left printed profile blocks all three; the right has stable AB/CD. Thus the complete strict two-side existence theorem cannot be applied unchanged to one-group roommates.

Children investigate score versus stability, omitted choices, and one-group existence: three directions. Grades 2-3 can build/check visible pairs; Grades 4-5 can certify optima, nonexistence and the matched-set invariant. The real gate is retaining current partners and checking both wishes from one unchanged state. Search supplies examples; a complete blocker catalog or alternating-path contradiction supplies an explanation.

Entry routes and readiness

Fixed tables: K–1 / 2–3 / 4–5, with one anchored adult each. Use one investigation and its brief launch per table visit. Adults read/record while children own the mathematical choices. Entry needs short ordered lists, fixed current partners and both preference checks. No separate K-1 route is forced; an oral four-token trial is optional only after this gate is retained.

P1 adds rank sums through 15; P2 adds the unpaired state. P3 uses one group, not circles paired only with squares. P4's universal explanation is readiness-dependent return work.

Materials and preparation

Per table: labeled artificial tokens A,B,C,X,Y,Z; six reusable preference strips and six blanks; removable pairing links; rank cards 1-3; visible unpaired spaces; pencils and paper. For P3 supply four same-style tokens A-D and four strips. Large labels about 20 mm help pointing. Do not rank real children's friendships or desirability.

Prepare the printed lists and preserve omissions genuinely; a missing choice is not tied last. Allowed edges require both names. Keep current pairs and preferences unchanged while checking blockers, and keep unpaired objects visible. About 10-15 minutes to prepare reusable strips; independent child retention of the simultaneous checks remains untested.

Launch together

After handling tokens, use the final visual: current UP and VQ, V's list P,Q and P's list V,U, with current partners boxed. V prefers P to Q and P prefers V to U, so VP blocks. Do not change a pair between the two checks. For P2 show an unacceptable omitted edge; for P3 explicitly remove the two-side rule.

First visit and return visits

Allow a 2-3 minute launch and 20-35 minutes for P1 or P2. Return for P3's contrasting existence cases or P4's general invariant. These timings are proposals. Retain a list profile and a checked reference matching; allow children to stay with one investigation instead of finishing all four pages.

Problem 1: Lowest score is not always stable

Entry, prerequisites and timing

Build candidates and attach their two rank numbers to each pair. Grades 2-3 can pool the six cases; older children can certify both minima. Allow 20-30 minutes. Rank score and a blocker are separate records about the same unchanged matching.

Mathematical facts and solutions

All six perfect matchings, totals and full blockers: AX/BY/CZ: 15, AY/AZ/CY; AX/BZ/CY: 13, AY/BX; AY/BX/CZ: 11, none; AY/BZ/CX: 10, BX; AZ/BX/CY: 11, AY; AZ/BY/CX: 12, AY. Fixing A's partner then matching the two others gives exactly six, so this is complete.

The unique aggregate minimum AY/BZ/CX scores 10. B ranks X second ahead of current Z third, and X ranks B first ahead of current C third: BX blocks. The unique stable result AY/BX/CZ has pair scores 2,3,6 totaling 11, hence is also the lowest stable total. Maximizing a different notion of collective benefit is not what stability alone guarantees.

Hold-back hints

Let a child compare B/X's wishes in the lowest-score reference before checking every other potential pair. For completeness, offer movable matching links rather than prescribing a written enumeration order.

Depth and further questions

Why can a low aggregate total conceal a mutually preferred switch? In the printed minimum AY/BZ/CX, forming BX breaks BZ and CX, leaving C and Z unpaired until re-paired; AY stays. Pairing CZ completes AY/BX/CZ. A blocker test and a complete rearrangement are different questions. Keep artificial scores and real preferences conceptually separate.

Facilitator notes

A unilateral wish is not enough. Adult reading support can help; adults making all blocker decisions would remove the investigation.

Problem 2,4: Partners can change; matched objects cannot

Entry, prerequisites and timing

P2 uses small omitted-choice profiles and visible unpaired objects. P4 lets children design lists and seek a counterexample. Allow 20-40 minutes; the general proof can be a separate upper return conversation using alternating colored links.

Mathematical facts and solutions

P2 first profile: AX is uniquely stable, leaving B and Y unpaired. Empty matching is blocked by AX; BX leaves A unpaired and is blocked by AX because X prefers A. Y has no allowed edge. Equal side sizes do not force everyone to pair.

Second profile: exactly AX/BY and AY/BX are stable, both leaving C,Z unpaired. Z has no edge. If CY were used, one of A/B would be unpaired because only X remains; that free object and Y block, since Y ranks A and B ahead of C. Without C matched, neither A nor B can stay unpaired: a free acceptable X/Y, or the receiver's preference for the unpaired object over its current partner, creates a blocker. Both A/B full alternatives pass both-sided checks. Partners differ, unpaired objects do not.

P4's requested counterexample does not exist under the shared conventions. Suppose p0 is unpaired in stable M but paired with r0 in stable N. M-stability forces r0 to have an M-partner p1 preferred to p0. N-stability forces p1 to have an N-partner r1 preferred to r0, otherwise p1/r0 block N. M-stability now forces r1 to have an M-partner p2 preferred to p1, and so on.

The M/N links form an alternating path beginning at an unmatched endpoint. It cannot repeat: a matching gives at most one edge of each type, so any repeated alternating component would be a cycle disconnected from that endpoint. It cannot terminate without the blocking pair just forced. Finiteness contradicts endless continuation. Exchange M/N and then the sides to prove equality of matched sets. Strict preferences, mutual acceptability and acceptable-above-unpaired are essential; failed counterexample search alone is not this proof.

Hold-back hints

Keep unpaired tokens in view. When CY is tried, ask which object now has no partner and whether Y prefers it. For P4, preserve two proposed stable matchings side by side and follow alternating links from an object claimed to change status.

Depth and further questions

Starting from the first profile, remove A-X from both lists. BX is now uniquely stable, leaving A,Y unpaired. Changing the profile can change unmatched objects; the theorem compares matchings of one fixed profile, not different profiles.

Facilitator notes

Zero-choice objects remain participants. An omitted acceptable relation must be absent from the allowed-edge set; do not use an unacceptable pair to fill an empty place.

Problem 3: One group may have no stable pairing

Entry, prerequisites and timing

Four identical-style tokens permit every within-group pair. Keep one candidate matching fixed while testing both wishes. Allow 15-30 minutes. A child can explain nonexistence by exhausting three drawings, without a matching algorithm.

Mathematical facts and solutions

There are exactly three perfect pairings: choose A's partner B,C or D, and the remaining two must pair. In the left profile AB/CD is blocked by BC; AC/BD by AB; AD/BC by AC. Every candidate therefore fails, proving no stable perfect pairing exists.

In the right profile AB/CD consists of two mutual first-choice pairs and is stable. AC/BD is blocked by AB and CD. AD/BC is blocked by AB, AC and CD. Thus AB/CD is uniquely stable. Some one-group profiles work and others do not; the left example refutes a universal existence claim for this changed model.

The base deferred-acceptance theorem assumes two disjoint sides with every pair crossing sides. It does not imply a stable solution for this one-group question. Here perfect pairing is required; leaving everyone unpaired is not a permitted answer.

Hold-back hints

After a genuine attempt, fix A-B and check the remaining possible mutual wishes. Use the three matching boards to establish completeness; do not interpret an algorithm's failure as a nonexistence proof.

Depth and further questions

Any one-group profile with all objects organized into mutual first-choice pairs has a stable matching of those pairs: no object prefers anyone else. The right profile illustrates that sufficient construction.

Facilitator notes

Use fixed artificial labels and the same preference strips throughout an instance. Repeatedly updating partners while testing blockers changes the reference and can hide the intended contradiction.

Sources, checks and classroom record

Sources and adaptation

Current Week 27 base packets/guide establish strict complete two-side stability. The base outline cites Pass and Halpern, A Course in Networks and Markets, Chapter 14 sections 14.1-14.3 for that model. No new source-body reading or attribution of these new examples is claimed. The score and roommate catalogs and incomplete-list proof are independently supplied here.

Rozhkovskaya, Math Circles for Elementary School Students, Introduction: Berkeley 2009 (OEBPS/part0010.xhtml), actually reports individual adult support. Our visible strips and anchored adult help are proposed adaptations. Week 13 allowed-edge capacity is related but lacks these preference/blocker conditions.

Verification

The fresh student math critic recomputed every score/blocker, both partial-matching profiles and both roommate profiles, and independently checked incomplete-list examples plus the general alternating-path proof. Final revision supplied the actual ordered input strips in the blocking visual; guide answers use the revised final packet.

The final guide received a separate independent mathematical review. Both final PDFs were rendered and inspected and passed a clean extracted-source ZIP rebuild. See release-validation.json in the editable source package. No physical procedure, staffing success or independent child retention has been rehearsed. All materials are unpiloted.

What to record after a visit

Record investigations used, actual preference profiles, retained matchings and a useful continuation.

Observe whether both checks are made from one unchanged reference, missing choices remain unacceptable, unpaired tokens stay visible, and a child controls the blocker decision. Separate those observations from readiness hypotheses.

Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.